Algorithm 1 Two-Step pp-ADMM-NM Algorithm for Spacing Optimization

Input: Initial simplex $\{d_s^{(n_s)}\}_{n_s=1}^{N_s+1}$ with $d_s^{(n_s)} \in [d_{\min}, d_{\max}]$, pp-ADMM tolerance ε, NM coefficients $(\alpha, \gamma, \rho, \sigma)$, simplex tolerance τ.

Set iteration counter $q = 0$.
repeat
  Step 1 - pp-ADMM updates:
  for $n_s = 1$ to $N_s + 1$ **do**
    Set spacing $d_s = d_s^{(n_s)}$ and initialize pp-ADMM variables $\bar{\mathbf{B}}^0$, $\bar{\mathbf{U}}^0$, $\mathbf{W}^0$, $\mathbf{y}^0$ and $\boldsymbol{\gamma}^0$ [9, Section III].
    Iterate the pp-ADMM updates until convergence tolerance ε is reached.
    Compute the objective value $f(d_s^{(n_s)})$ using (13a) with the obtained $\mathbf{W}$ and $\bar{\mathbf{U}}$.
  end for
  Step 2 - NM update:
  Sort the vertices so that $f(d_s^{(1)}) \leq f(d_s^{(2)}) \leq \cdots \leq f(d_s^{(N_s+1)})$.
  Compute the centroid $d_{s,c}^{(q)}$ of the best N_s points using (14).
  Reflection: $d_{s,r}^{(q)} = d_{s,c}^{(q)} + \alpha\,(d_{s,c}^{(q)} - d_{s,N_s+1}^{(q)})$; evaluate $f(d_{s,r}^{(q)})$ via a pp-ADMM run.
  if $f(d_{s,1}^{(q)}) \leq f(d_{s,r}^{(q)}) < f(d_{s,N_s}^{(q)})$ **then**
    Replace $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,r}^{(q)}$.
  else if $f(d_{s,r}^{(q)}) < f(d_{s,1}^{(q)})$ **then**
    Expansion: $d_{s,e}^{(q)} = d_{s,c}^{(q)} + \gamma\,(d_{s,r}^{(q)} - d_{s,c}^{(q)})$; evaluate $f(d_{s,e}^{(q)})$.
    if $f(d_{s,e}^{(q)}) < f(d_{s,r}^{(q)})$ **then**
      $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,e}^{(q)}$.
    else
      $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,r}^{(q)}$.
    end if
  else if $f(d_{s,N_s}^{(q)}) \leq f(d_{s,r}^{(q)}) < f(d_{s,N_s+1}^{(q)})$ **then**
    Outside contraction: $d_{s,\mathrm{oc}}^{(q)} \leftarrow d_{s,c}^{(q)} + \rho\,(d_{s,r}^{(q)} - d_{s,c}^{(q)})$; evaluate $f(d_{s,\mathrm{oc}}^{(q)})$.
    if $f(d_{s,\mathrm{oc}}^{(q)}) \leq f(d_{s,r}^{(q)})$ **then**
      Replace $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,\mathrm{oc}}^{(q)}$.
    else
      Shrinkage: For $i = 2, \ldots, N_s + 1$ set $d_{s,i}^{(q)} \leftarrow d_{s,1}^{(q)} + \sigma\,(d_{s,i}^{(q)} - d_{s,1}^{(q)})$ and re-evaluate $f(d_{s,i}^{(q)})$ via pp-ADMM.
    end if
  else
    Inside contraction: $d_{s,\mathrm{ic}}^{(q)} \leftarrow d_{s,c}^{(q)} - \rho\,(d_{s,r}^{(q)} - d_{s,c}^{(q)})$; evaluate $f(d_{s,\mathrm{ic}}^{(q)})$.
    if $f(d_{s,\mathrm{ic}}^{(q)}) < f(d_{s,N_s+1}^{(q)})$ **then**
      Replace $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,\mathrm{ic}}^{(q)}$.
    else
      Shrinkage: For $i = 2, \ldots, N_s + 1$ set $d_{s,i}^{(q)} \leftarrow d_{s,1}^{(q)} + \sigma\,(d_{s,i}^{(q)} - d_{s,1}^{(q)})$ and re-evaluate $f(d_{s,i}^{(q)})$ via pp-ADMM.
    end if
  end if
  $q \leftarrow q + 1$.
until $\max(\mathcal{D}_s^{(q)}) \leq \tau$

Output: Best spacing $d_s^\star = d_{s,1}^{(q)}$ and corresponding $\mathbf{W}^\star$, $\bar{\mathbf{B}}^\star$ and $\bar{\mathbf{U}}^\star$ obtained by pp-ADMM.